Qing Qu

EECS Department,
University of Michigan - Ann Arbor,
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Research Interest

High-dimensional Data Analysis, Theory of Deep Learning, Machine Learning for Scientific Discovery

PROFESSIONAL EXPERIENCE

Assistant Professor Jan. 2021 – present
ECE division of EECS Dept., University of Michigan - Ann Arbor, MI, USA
Moore-Sloan Research Fellow Dec. 2018 – Dec. 2020
Center for Data Science, New York University, NY, USA
research assistant professor with limited term

EDUCATION

Ph.D. Electrical Engineering Oct. 2018
Columbia University in the City of New York, New York, USA
Thesis: Nonconvex recovery of low-dimensional models [link], Advisor: Prof. John N. Wright

M.S.E. Electrical & Computer Engineering May 2013
Johns Hopkins University, Baltimore, USA
Research focus: compressed sensing and sparse representations, Advisor: Prof. Trac D. Tran

B.E. Electronic Engineering Jul. 2011
Tsinghua University, Beijing, China

HONORS AND AWARDS

Best Paper Award, NeurIPS Workshop on Diffusion Models 2023
Amazon Research Awards, AWS AI 2023
NSF Career Awards 2022
National Science Foundation's most prestigious awards in support of early-career faculty
Top Reviewer for NeurIPS & ICML 2018-2021
Recognized as an outstanding reviewer (top 8%) for NeurIPS'21, top 33% reviewer for ICML'20, top 400 (top 10%) reviewer for NeurIPS'19, and top 30% reviewer for NeurIPS'18
NYU Moore-Sloan Data Science Fellowship 2018 – 2020
Microsoft Research Ph.D. Fellowship Sept. 2016 – May 2018
Only 12 Ph.D. graduate students in EECS across North Amercia were awarded annually.
Best Student Paper Award in SPARS'15 2015
Joint with Ju Sun and John Wright
Dean Robert H. Roy Fellowship 2011
Prestigious fellowship from the Whiting School of Engineering, the Johns Hopkins University
First Class Award for Student Research Training 2010
Awarded for top 3 of over 700 undergraduate research projects across school, Tsinghua University
Project: sparse signal recovery for compressed sensing, Advisor: Prof. Yuantao Gu

REPRESENTATIVE PUBLICATIONS

Total citations: 3389, H-index: 27, according to Google Scholar as of February 2, 2025. Please refer to my Google Scholar for updated citations. † indicates equal contributions.

[1] Huijie Zhang†, Jinfan Zhou†, Yifu Lu, Minzhe Guo, Liyue Shen, and **Qing Qu**. The Emergence of Reproducibility and Consistency in Diffusion Models. In *International Conference on Machine Learning (ICML)*, 2024 (**Best Paper Award** of 2023 NeurIPS Workshop on Diffusion Models).

[2] Can Yaras, Peng Wang, Laura Balzano, and **Qing Qu**. Compressible Dynamics in Deep Overparameterized Low-Rank Learning & Adaptation. In *International Conference on Machine Learning (ICML)*, 2024 (**Oral, top 1.5%**, **Best Poster Award** at MMLS'24).

[3] Bowen Song, Soo-Min Kwon, Zecheng Zhang, Xinyu Hu, **Qing Qu**, and Liyue Shen. Solving Inverse Problems with Latent Diffusion Models via Hard Data Consistency. In *International Conference on Learning Representations (ICLR)*, 2024 (**Spotlight, top 4%**).

[4] Zhihui Zhu†, Tianyu Ding†, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and **Qing Qu**. A Geometric Analysis of Neural Collapse with Unconstrained Features. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021 (**Spotlight, top 3%**).

[5] **Qing Qu**, Yuexiang Zhai, Xiao Li, Yuqian Zhang, and Zhihui Zhu. Geometric Analysis of Nonconvex Optimization Landscapes for Overcomplete Learning. In *International Conference on Learning Representations (ICLR)*, 2020 (**Oral, the highest review score, top 1.9%**).

[6] Chong You†, Zhihui Zhu†, **Qing Qu**, and Yi Ma. Robust Recovery via Implicit Bias of Discrepant Learning Rates for Double Over-parameterization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020 (**Spotlight, top 4%**).

[7] **Qing Qu**, Xiao Li, and Zhihui Zhu. Exact Recovery of Multichannel Sparse Blind Deconvolution via gradient descent. *SIAM Journal on Imaging Sciences*, 13(3):1630–1652, 2021 (*NeurIPS'19*, **Spotlight, top 3%**).

[8] Ju Sun, **Qing Qu**, and John Wright. A Geometric Analysis of Phase Retrieval. *Foundations of Computational Mathematics*, 18(5):1131–1198, 2018 (**Frontiers of Science Award** in Mathematics, ICBS'24).

[9] Ju Sun, **Qing Qu**, and John Wright. Complete Dictionary Recovery using Nonconvex Optimization. In *International Conference on Machine Learning (ICML)*, pages 2351–2360, 2015 (**Best Paper Award** at SPARS'15).

GRANTS

Total funding raised: $6,089,280; PI's own share: $3,286,938

Current Projects

Army Research Office (ARO) W911NF-25-1-0047 2025
Efficient and Robust Learning for Domain Shift and Data Scarcity at the Edge. Award Amount: $300,000, Period Covered: 3/01/2025 - 02/28/2027
PI: **Qing Qu**, Co-PI: JJ (Jeong Joon) Park (PI's share: $150,000)

NSF CISE Medium, Award #2402950 2024
Collaborative Research: RI: Medium: Principled Approaches to Deep Learning for Low-dimensional Structures Award Amount: $1,200,000, Period Covered: 7/15/2024 - 06/30/2027
PI: **Qing Qu**, Co-PI: Zhihui Zhu, Co-PI: Yi Ma (PI's share: $400,000)

NSF CISE Medium, Award #2312842 2023
Collaborative Research: RI: Medium: Principles for Optimization, Generalization, and Transferability via Deep Neural Collapse Award Amount: $1,200,000, Period Covered: 10/01/2023 - 09/30/2026
PI: Zhihui Zhu, Co-PI: Jeremias Sulam, Co-PI: **Qing Qu** (PI's share: $400,000)

Amazon Research Awards, AWS AI 2023

Single PI, Award Amount: $70,000 gift fund + $50,000 AWS credit

KLA Corporation Gift Grant 2023-2024
Domain Adaptation via Machine Learning Single PI, Award Amount: $100,000

MICDE Catalyst Grant 2023
Efficient Diffusion Models for Scientific Machine Learning Award Amount: $100,000, Period Covered: 11/01/2023 - 10/31/2024
PI: Liyue Shen, Co-PI: Jeff Fessler, Co-PI: **Qing Qu**

NSF Career, CCF, Award #2143904, Single PI: Qing Qu 2022
From Shallow to Deep Representation Learning: Global Nonconvex Optimization Theories and Efficient Algorithms Award Amount: $633,280 + AWS Credit $15,000, Period Covered: 05/01/2022 - 05/01/2027

NSF CISE Medium, Award #2212066 2022
Collaborative Research: CIF: Medium: Taming Deep Unsupervised Representation Learning in Imaging: Theory and Algorithms Award Amount: $1,200,000, Period Covered: 10/01/2022 - 09/30/2025
PI: Saiprasad Ravishankar, Co-PI: Matthew Hirn, Co-PI: **Qing Qu** (PI's share: $370,703 + REU $16,000 + AWS Credit $15,000)

NSF CISE Medium, Award #2212326 2022
Collaborative Research: CIF: Medium: Foundations of Robust Deep Learning via Data Geometry and Dyadic Structure Award Amount: $1,200,000, Period Covered: 10/01/2022 - 09/30/2026
PI: Kevin Moon, Co-PI: Abram Magner, Co-PI: **Qing Qu** (transferred from PI Alfred Hero, PI's share: $471,955 + AWS Credit $15,000)

Office of Naval Research (ONR), N00014-22-S-B001, Single PI: Qing Qu 2022
Towards Efficient (Deep) Representation Learning Through Nonconvex Optimization and Low-dimensional Models Award Amount: $360,000, Period Covered: 07/01/2022 - 06/30/2025

Completed Projects

Seeding To Accelerate Research Themes (START) Grant, University of Michigan 2022
Next Generation Data Science: from High Dimensional Statistics to Nonconvex Optimization Budget: $75,000, Period Covered: 07/01/2022 - 06/30/2023
PI: **Qing Qu** (PI's share: $15,000), Co-PI: Laura Balzano, Co-PI: Salar Fatthi, Co-PI: Albert Berahas, Co-PI: Eunshin Byon

Propelling Original Data Science (PODS) Grant, University of Michigan 2022
Machine Learning Guided Co-design for Reconstructive Spectroscopy Budget: $50,000, Period Covered: 08/01/2022 - 07/30/2023
PI: **Qing Qu** (PI's share: $30,000), Co-PI: Pei-Cheng Ku

NSF DMS, Award #2009752 2020
Mathematical Analysis of Super-resolution via Nonconvex Optimization and Machine Learning
Award Amount: $ 340,000, Period Covered: 08/01/2020 - 07/31/2024
PI: Carlos Fernandez-Granda, Co-PI: **Qing Qu** (transferred to the PI on 12/31/2020)

FULL PUBLICATION LIST

Preprints

[1] Xiao Li, Zekai Zhang, Xiang Li, Siyi Chen, Zhihui Zhu, Peng Wang, and **Qing Qu**. Understanding Inner Representation of Diffusion Models via Low-Dimensional Modeling. *In submission to ICML'25*, 2025.

[2] Siyi Chen, Yimeng Zhang, Sijia Liu, and **Qing Qu**. The Dual Power of Interpretable Token Embeddings: Jailbreaking Attacks and Defenses for Diffusion Model Unlearning. *In submission to ICML'25*, 2025.

[3] Xiang Li, Yixiang Dai, Rongrong Wang, and **Qing Qu**. Towards Understanding the Mechanisms of Classifier-Free Guidance. *In submission to ICML'25*, 2025.

[4] Peng Wang, Yifu Lu, Yaodong Yu, Druv Pai, **Qing Qu**, and Yi Ma. Attention-Only Transformers via Unrolled Subspace Denoising. *In submission to ICML'25*, 2025.

[5] Alec S Xu, Can Yaras, Peng Wang, and **Qing Qu**. Understanding How Nonlinear Layers Create Linearly Separable Features for Low-Dimensional Data. *ArXiv Preprint arXiv:2501.02364, in submission to SIAM Journal on Mathematics of Data Science*, 2025.

[6] Siyi Chen†, Yixuan Jia†, **Qing Qu**, He Sun, and Jeffrey A Fessler. FlowDAS: A Flow-Based Framework for Data Assimilation. *ArXiv Preprint arXiv:2501.16642, in submission to CVPR'25*, 2025.

[7] Peng Wang†, Huijie Zhang†, Zekai Zhang, Siyi Chen, Yi Ma, and **Qing Qu**. Diffusion Model Learns Low-Dimensional Distributions via Subspace Clustering. *Arxiv Preprint arXiv:2409.02426, in submission to the Proceedings of the National Academy of Sciences (PNAS)*, 2025.

[8] Can Yaras†, Siyi Chen†, Peng Wang, and **Qing Qu**. Explaining and Mitigating the Modality Gap in Contrastive Multimodal Learning. *ArXiv Preprint arXiv:2412.07909, in submission to CPAL'25*, 2024.

[9] Huminhao Zhu, Fangyikang Wang, Tianyu Ding, **Qing Qu**, and Zhihui Zhu. Analyzing and Improving Model Collapse in Rectified Flow Models. *ArXiv Preprint arXiv:2412.08175, in submission to ICML'25*, 2024.

[10] Wenda Li†, Huijie Zhang†, and **Qing Qu**. Shallow Diffuse: Robust and Invisible Watermarking through Low-Dimensional Subspaces in Diffusion Models. *ArXiv Preprint arXiv:2410.21088, in submission to ICML'25*, 2024.

[11] Ismail Alkhouri, Shijun Liang, Cheng-Han Huang, Jimmy Dai, **Qing Qu**, Saiprasad Ravishankar, and Rongrong Wang. SITCOM: Step-wise Triple-Consistent Diffusion Sampling for Inverse Problems. *ArXiv Preprint arXiv:2410.04479, in submission to ICML'25*, 2024.

[12] Siyi Chen, Minkyu Choi, Zesen Zhao, Kuan Han, **Qing Qu**, and Zhongming Liu. Unfolding Videos Dynamics via Taylor Expansion. *Arxiv Preprint arXiv:2409.02371, in submission to CPAL'25*, 2024.

[13] Xiang Li, Soo Min Kwon, Ismail Alkhouri, Saiprasad Ravishankar, and **Qing Qu**. Decoupled Data Consistency with Diffusion Purification for Image Restoration. *ArXiv Preprint arXiv:2403.06054, in submission to ICCV'25*, 2024.

[14] Peng Wang†, Xiao Li†, Can Yaras, Zhihui Zhu, Laura Balzano, Wei Hu, and **Qing Qu**. Understanding Deep Representation Learning via Layerwise Feature Compression and Discrimination. *ArXiv Preprint arXiv:2311.02960, in submission to Journal of Machine Learning Research (JMLR)*, 2023.

[15] Can Yaras†, Peng Wang†, Wei Hu, Zhihui Zhu, Laura Balzano, and **Qing Qu**. The Law of Parsimony in Gradient Descent for Learning Deep Linear Networks. *ArXiv Preprint arXiv:2306.01154*, 2023.

[16] Ismail Alkhouri, Shijun Liang, Rongrong Wang, **Qing Qu**, and Saiprasad Ravishankar. Diffusion-based Adversarial Purification for Robust Deep MRI Reconstruction. *ArXiv Preprint arXiv:2309.05794, in submission to IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2024.

[17] Yuqian Zhang, **Qing Qu**, and John Wright. From Symmetry to Geometry: Tractable Nonconvex Problems. *ArXiv Preprint arXiv:2007.06753*, 2022.

[18] **Qing Qu**†, Zhihui Zhu†, Xiao Li, M. C. Tsakiris, John Wright, and René Vidal. Finding the Sparsest Vectors in a Subspace: Theory, Algorithms, and Applications. *ArXiv Preprint arXiv:2001.06970*, 2021.

Conference Proceedings

[1] Avrajit Ghosh, Soo Min Kwon, Rongrong Wang, Saiprasad Ravishankar, and **Qing Qu**. Learning Dynamics of Deep Matrix Factorization Beyond the Edge of Stability. In *International Conference on Learning Representations (ICLR)*, 2025.

[2] Siyi Chen†, Huijie Zhang†, Minzhe Guo, Yifu Lu, Peng Wang, and **Qing Qu**. Exploring Low-Dimensional Subspace in Diffusion Models for Controllable Image Editing. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[3] Xiang Li, Yixiang Dai, and **Qing Qu**. Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] Ismail Alkhouri, Shijun Liang, Evan Bell, **Qing Qu**, Rongrong Wang, and Saiprasad Ravishankar. Image Reconstruction via Autoencoding Sequential Deep Image Prior. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[5] Changwoo Lee, Soo Min Kwon, **Qing Qu**, and Hun-Seok Kim. BLAST: Block-Level Adaptive Structured Matrices for Efficient Deep Neural Network Inference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[6] Huijie Zhang†, Jinfan Zhou†, Yifu Lu, Minzhe Guo, Liyue Shen, and **Qing Qu**. The Emergence of Reproducibility and Consistency in Diffusion Models. In *International Conference on Machine Learning (ICML)*, 2024 (**best paper award** of 2023 NeurIPS Workshop on Diffusion Models).

[7] Can Yaras, Peng Wang, Laura Balzano, and **Qing Qu**. Compressible Dynamics in Deep Overparameterized Low-Rank Learning & Adaptation. In *International Conference on Machine Learning (ICML)*, 2024 (**oral, top 1.5%**).

[8] Pengyu Li†, Yutong Wang†, Xiao Li, and **Qing Qu**. Neural Collapse in Multi-label Learning with Pick-all-label Loss. In *International Conference on Machine Learning (ICML)*, 2024.

[9] Jiachen Jiang, Jinxin Zhou, Peng Wang, **Qing Qu**, Dustin Mixon, Chong You, and Zhihui Zhu. Generalized Neural Collapse for a Large Number of Classes. In *International Conference on Machine Learning (ICML)*, 2024.

[10] Peng Wang, Huikang Liu, Druv Pai, Yaodong Yu, Zhihui Zhu, **Qing Qu**, and Yi Ma. A Global Geometric Analysis of Maximal Coding Rate Reduction. In *International Conference on Machine Learning (ICML)*, 2024.

[11] Huikang Liu, Peng Wang, Longxiu Huang, **Qing Qu**, and Laura Balzano. Matrix Completion with ReLU Sampling. In *International Conference on Machine Learning (ICML)*, 2024.

[12] Avrajit Ghosh, Xitong Zhang, Kenneth K. Sun, **Qing Qu**, Saiprasad Ravishankar, and Rongrong Wang. Optimal Eye Surgeon: Finding Image Priors through Sparse Generators at Initialization. In *International Conference on Machine Learning (ICML)*, 2024.

[13] Soo Min Kwon†, Zekai Zhang†, Dogyoon Song, Laura Balzano, and **Qing Qu**. Efficient Low-Dimensional Compression of Overparameterized Models. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 1009–1017, 2024.

[14] Huijie Zhang†, Yifu Lu†, Ismail Alkhouri, Saiprasad Ravishankar, Dogyoon Song, and **Qing Qu**. Improving Training Efficiency of Diffusion Models via Multi-Stage Framework and Tailored Multi-Decoder Architectures. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[15] Bowen Song, Soo-Min Kwon, Zecheng Zhang, Xinyu Hu, **Qing Qu**, and Liyue Shen. Solving Inverse Problems with Latent Diffusion Models via Hard Data Consistency. In *International Conference on Learning Representations (ICLR)*, 2024 (**spotlight, top 4%**).

[16] Yuexiang Zhai, Shengbang Tong, Xiao Li, Mu Cai, **Qing Qu**, Yong Jae Lee, and Yi Ma. Investigating the Catastrophic Forgetting in Multimodal Large Language Models. In *Conference on Parsimony and Learning (CPAL)*, 2024 (**oral**).

[17] Evan Bell, Shijun Liang, **Qing Qu**, and Saiprasad Ravishankar. Robust Self-Guided Deep Image Prior. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2023 (**oral**).

[18] Can Yaras†, Peng Wang†, Zhihui Zhu, Laura Balzano, and **Qing Qu**. Neural Collapse with Normalized Features: A Geometric Analysis over the Riemannian Manifold. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[19] Jinxin Zhou, Chong You, Xiao Li, Kangning Liu, Liu Sheng, **Qing Qu**, and Zhihui Zhu. Are All Losses Created Equal: A Neural Collapse Perspective. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[20] Peng Wang†, Huikang Liu†, Can Yaras†, Laura Balzano, and **Qing Qu**. Linear Convergence Analysis of Neural Collapse with Unconstrained Features. In *NeurIPS'22 Workshop: Optimization for Machine Learning (In Preparation for SIAM Journal on Optimization)*, 2022.

[21] Shuo Xie, Jiahao Qiu, Ankita Pasad, **Qing Qu**, and Hongyuan Mei. Hidden State Variability of Pretrained Language Models Can Guide Computation Reduction for Transfer Learning. In *Findings of the Association for Computational Linguistics: EMNLP*, 2022.

[22] Sheng Liu, Zhihui Zhu, **Qing Qu**, and Chong You. Robust Training under Label Noise by Over-parameterization. In *International Conference on Machine Learning (ICML)*, 2022.

[23] Jinxin Zhou†, Xiao Li†, Tianyu Ding, Chong You, **Qing Qu**†, and Zhihui Zhu†. On the Optimization Landscape of Neural Collapse under MSE Loss: Global Optimality with Unconstrained Features. In *International Conference on Machine Learning (ICML)*, 2022.

[24] Lijun Ding†, Liwei Jiang†, Yudong Chen, **Qing Qu**, and Zhihui Zhu. Rank Overspecified Robust Matrix Recovery: Subgradient Method and Exact Recovery. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[25] Zhihui Zhu†, Tianyu Ding†, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and **Qing Qu**. A Geometric Analysis of Neural Collapse with Unconstrained Features. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021 (**spotlight, top 3%**).

[26] Sheng Liu†, Xiao Li†, Chong You, Zhihui Zhu, Carlos Fernandez-Granda, and **Qing Qu**. Convolutional Normalization: Improving Deep Convolutional Network Robustness and Training. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[27] Chong You†, Zhihui Zhu†, **Qing Qu**, and Yi Ma. Robust Recovery via Implicit Bias of Discrepant Learning Rates for Double Over-parameterization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020 (**spotlight, top 4%**).

[28] **Qing Qu**, Yuexiang Zhai, Xiao Li, Yuqian Zhang, and Zhihui Zhu. Geometric Analysis of Nonconvex Optimization Landscapes for Overcomplete Learning. In *International Conference on Learning Representations (ICLR)*, 2020 (**oral, the highest review score, top 1.9%**).

[29] Yenson Lau†, **Qing Qu**†, Han-Wen Kuo, Pengcheng Zhou, Yuqian Zhang, and John Wright. Short and Sparse Deconvolution — A Geometric Approach. In *International Conference on Learning Representations (ICLR)*, 2020.

[30] **Qing Qu**, Xiao Li, and Zhihui Zhu. A Nonconvex Approach for Exact and Efficient Multichannel Sparse Blind Deconvolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019 (**spotlight, top 3%**).

[31] **Qing Qu**, Yuqian Zhang, Yonina Eldar, and John Wright. Convolutional Phase Retrieval. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 6086–6096, 2017.

[32] Ju Sun, **Qing Qu**, and John Wright. A Geometric Analysis of Phase Retrieval. In *2016 IEEE International Symposium on Information Theory (ISIT)*, pages 2379–2383. IEEE, 2016.

[33] Ju Sun, **Qing Qu**, and John Wright. Complete Dictionary Recovery Over the Sphere. In *2015 International Conference on Sampling Theory and Applications (SampTA)*, 2015.

[34] **Qing Qu**, Ju Sun, and John Wright. Finding a Sparse Vector in a Subspace: Linear Sparsity Using Alternating Directions. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3401–3409, 2014.

[35] **Qing Qu**, Xiaoxia Sun, Nasser M. Nasrabadi, and Trac D. Tran. Subspace Vertex Pursuit for Separable Non-negative Matrix Factorization in Hyperspectral Unmixing. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8115–8119. IEEE, 2014.

[36] **Qing Qu**, Nasser M. Nasrabadi, and Trac D. Tran. Hyperspectral Abundance Estimation for the Generalized Bilinear Model with Joint Sparsity Constraint. In *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 2129–2133. IEEE, 2013.

Journals

[1] Shijun Liang, Evan Bell, **Qing Qu**, Rongrong Wang, and Saiprasad Ravishankar. Analysis of Deep Image Prior and Exploiting Self-Guidance for Image Reconstruction. *IEEE Transactions on Computational Imaging*, 2025.

[2] Jiyi Chen†, Pengyu Li†, Yutong Wang, Pei-Cheng Ku, and **Qing Qu**. Sim2Real in Reconstructive Spectroscopy: Deep Learning with Augmented Device-Informed Data Simulation. *APL Machine Learning*, 2(3):036106, August 2024.

[3] Xiao Li†, Sheng Liu†, Jinxin Zhou, Xinyu Lu, Carlos Fernandez-Granda, Zhihui Zhu, and **Qing Qu**. Understanding and Improving Transfer Learning of Deep Models via Neural Collapse. *Transactions on Machine Learning Research*, 2024.

[4] Tuba Sarwar, Can Yaras, Xiang Li, **Qing Qu**, and Pei-Cheng Ku. Miniaturizing a Chip-Scale Spectrometer Using Local Strain Engineering and Total-Variation Regularized Reconstruction. *Nano Letters*, 22(20):8174–8180, 2022.

[5] Xiao Li†, Shixiang Chen†, Zengde Deng, **Qing Qu**, Zhihui Zhu, and Anthony Man Cho So. Weakly Convex Optimization over Stiefel Manifold Using Riemannian Subgradient-Type Methods. *SIAM Journal on Optimization*, 31(3):1605–1634, 2021.

[6] **Qing Qu**, Xiao Li, and Zhihui Zhu. Exact Recovery of Multichannel Sparse Blind Deconvolution via gradient descent. *SIAM Journal on Imaging Sciences*, 13(3):1630–1652, 2021.

[7] **Qing Qu**, Yuqian Zhang, Yonina C. Eldar, and John Wright. Convolutional Phase Retrieval via Gradient Descent. *IEEE Trans. on Information Theory*, 66(3):1785–1821, 2020.

[8] Ju Sun, **Qing Qu**, and John Wright. A Geometric Analysis of Phase Retrieval. *Foundations of Computational Mathematics*, 18(5):1131–1198, 2018.

[9] **Qing Qu**, Ju Sun, and John Wright. Finding a Sparse Vector in a Subspace: Linear Sparsity Using Alternating Directions. *IEEE Trans. on Information Theory*, 62(10):5855–5880, 2016.

[10] Ju Sun, **Qing Qu**, and John Wright. Complete Dictionary Recovery Over the Sphere I: Overview and the Geometric Picture. *IEEE Trans. on Information Theory*, 63(2):853–884, 2016.

[11] Ju Sun, **Qing Qu**, and John Wright. Complete Dictionary Recovery Over the Sphere II: Recovery by Riemannian Trust-Region Method. *IEEE Trans. on Information Theory*, 63(2):885–914, 2016.

[12] **Qing Qu**, Nasser M. Nasrabadi, and Trac D. Tran. Subspace Vertex Pursuit: A Fast and Robust Near-Separable Nonnegative Matrix Factorization Method for Hyperspectral Unmixing. *IEEE Journal of Selected Topics in Signal Processing*, 9(6):1142–1155, 2015.

[13] **Qing Qu**, Nasser M. Nasrabadi, and Trac D. Tran. Abundance Estimation for Bilinear Mixture Models via Joint Sparse and Low-Rank Representation. *IEEE Trans. on Geoscience and Remote Sensing*, 52(7):4404–4423, 2013.

[14] Xiaoxia Sun, **Qing Qu**, Nasser M. Nasrabadi, and Trac D. Tran. Structured Priors for Sparse-Representation-Based Hyperspectral Image Classification. *IEEE Geoscience and Remote Sensing Letters*, 11(7):1235–1239, 2013.

[15] Jian Jin, **Qing Qu**, and Yuantao Gu. Robust Zero-Point Attraction Least Mean Square Algorithm on Near Sparse System Identification. *IET Signal Processing*, 7(3):210–218, 2013.

SELECTED PRESENTATIONS

The Emergence of Generalizability and Semantic Low-dim subspaces in Diffusion Models Jan. - Jan. 2025

Invited talks at UCSD HDSI Special Seminar, UC Berkeley Stats Seminar, ITA Workshop, KAIST, KLA Corporation, NeurIPS'23 Diffusion Model Workshop, KU IDS, UMich GenAI symposium, DeepMath Conference, Gatech ML Seminar

Invariant Low-Dimensional Subspaces in Gradient Descent for Learning Deep Networks Sept. 2023 - Apr. 2024

Invited talks at UMich CSP Seminar, Allerton'23, NYU MaD Seminar, UMN DSI Seminar, Columbia EE Seminar, UW Madison SILO Seminar, Young Scientist Forum at NUS, Google Research, UC Davis Math Seminar

Understanding Deep Representation Learning via Neural Collapse Sept. 2022 - Jun. 2023
Invited talks at Google Research NYC, Flatiron Institute CCM, SLowDNN Workshop, INFORMS'22, Allerton'22, UMich CSP Seminar, MSU, HKUST, CUHK

From Shallow to Deep Representation Learning Sept. 2021 - Dec. 2021
Invited seminar talks at IEEE SPS Webinar, University of Michigan Stats Seminar, Columbia University Stats Department, Princeton EE Department

Learning Low-Complexity Models from Data Feb. 2020 - Apr. 2020
Invited seminar talks at Michigan State U. CMSE, Northeastern U. ECE, University of Maryland College Park ECE, University of Michigan ECE, University of British Columbia ECE, University of Colordao Boulder ECEE, Penn. State U. EE, Dartmouth U. CS, University of Florida CISE & ECE

Sparse Deconvolution: Geometry, Methods & Applications Nov. 2019 - Jan. 2020
Invited talk at UC Berkeley EECS, Johns Hopkins University MINDS, NYU Center for Data Science, Tsinghua University EE, Xiamen University, NYU Tandon ECE

Nonconvex Optimization for Multichannel Sparse Blind Deconvolution Aug. - Dec. 2019
Invited talk in the Sixth International Conference on Continuous Optimization (ICCOPT).
Poster presentation at IMA Computational Imaging workshop, University of Minnesota.
Spotlight representation in Advances in Neural Information Processing Systems (NeurIPS).

Short-and-Sparse Deconvolution – A Geometric Approach Aug. - Oct. 2019
Contributed talk in the Sixth International Conference on Continuous Optimization (ICCOPT).
Poster presentation at IMA Computational Imaging workshop, University of Minnesota.

Nonconvex Recovery of Low-Complexity Models for Data Science Jan. 2019
Invited talk at NYU CDS lunch seminar, Purdue University ECE, JHU MINDS.

TUTORIALS & TEACHING

Leading Organizer & Instructor Aug. 2024
One week K12 Summer Camp: AI Magic
Co-taught with Changxiao Cai, JJ Park, and Liyue Shen

Leading Organizer & Instructor Jun. 2024
One-day tutorial at CVPR'24: Learning Deep Low-Dimensional Models from High-Dimensional Data: From Theory to Practice
Co-taught with Sam Buchanan, Yi Ma, Yuqian Zhang, and Zhihui Zhu

Leading Organizer & Instructor Apr. 2024
Half-day tutorial at ICASSP'24: Understanding Deep Representation Learning via Neural Collapse
joint presenting with Laura Balzano, Peng Wang, and Zhihui Zhu

Leading Organizer & Instructor 2022, 2023
Short Education Course at ICASSP'22 and ICASSP'23: Learning Nonlinear and Deep Low-Dimensional Representations from High-Dimensional Data: From Theory to Practice
Co-taught with Sam Buchanan, Yi Ma, Atlas Wang, John Wright, Yuqian Zhang, and Zhihui Zhu

Invited Instructor Jun. 2023
Deep learning international summer school: Advanced Course on Data Science & Machine Learning
Given 4 lectures to over 200 graduate students

Instructor University of Michigan
EECS 453: Principles of Machine Learning, Fall 2021-2024

Instructor University of Michigan
EECS 559: Optimization Methods for SIPML, Winter 2021-2024

Project Supervisor New York University
CDS-GA-1006: Capstone Project in Data Science, Fall 2019, Fall 2020
Leading four master students on a research project of data science.

Course Advisor New York University

CDS-GA 1009: Practical Training for Data Science, Fall 2019 - Fall 2020 (4 consecutive semesters)

Academic advisor for 10 master students for their data science internship in top tech companies in NYC.

PROFESSIONAL SERVICES

Conference & Workshop Organizations

Generative AI: From Theory to Scientific Applications Sept. 2024

Organizer of MIDAS Mini-Symposium, joint with Liyue Shen

Computational Imaging Workshop, IMSI, UChicago Aug. 2024

Co-Organizer, joint with Saiprasad Ravishankar, Ulugbek Kamilov, Rebecca Willett, Jong Chul Ye, and Brendt Wohlberg

Midwest Machine Learning Symposium 2024 (MMLS'24), University of Minnesota 2024

Co-Organizer, joint with Mingyi Hong, Ju Sun, Qiaomin Xie, Soumik Sarkar, and Elena Zheleva.

Conference on Parsimony and Learning (CPAL) 2023-present

Co-founder and Program Chair, joint with Yi Ma, Atlas Wang, Yuejie Chi, Zhihui Zhu, Gitta Kutyniok, Rene Vidal, Harry Shum, Gintare Karolina Dziugaite

Generative AI: Diffusion Models for Scientific Machine Learning Sept. 2023

Organizer of MIDAS Mini-Symposium, joint with Liyue Shen and Jeff Fessler

UMich ECE Communications and Signal Processing (CSP) Seminar Fall 2022 - present

Leading Organizer, joint with Lei Ying, Liyue Shen

Workshop on Seeking Low-dimensionality in Deep Neural Networks (SLowDNN) 2020-2023

Organizer, joint with Atlas Wang, Zhihui Zhu, Chong You, Jeremias Sulam, Yuejie Chi, Yi Ma

Mini-Symposium at SIAM Conference on Optimization (OP23) Jun. 2023

Tractable Non-Convex Problems in Low-Dimensional Learning, Co-organizer, joint Peng Wang and Laura Balzano

Mini-Symposium at SIAM Conference on Mathematics of Data Science (MDS22) Sept. 2022

Deep Learning with Low-Dimensional Models, Co-organizer, joint with Chong You and Zhihui Zhu

IEEE SPS Computational Imaging Webinar Series 2021-2022

Co-organizer, joint Saiprasad Ravishankar, Bihan Wen, Raja Giryes, Zhizhen Zhao, Jong Chul Ye

Affliations & Services

Member of IEEE Machine Learning and Signal Processing (MLSP) Technical Committee,
Member of IEEE Computational Imaging (CI) Technical Committee,

Conference Area Chair & Reviewer

Area Chair for ICASSP'22, AAAI'22, ICASSP'23, NeurIPS'23-24, ICLR'24-25, ICASSP'24, ICML'25

Reviewer for Conference on Learning Theory (COLT), Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), International Conference on Machine Learning (ICML), International Conference on Artificial Intelligence and Statistics (AISTATS), AAAI Conference on Artificial Intelligence (AAAI), International Symposium on Information Theory (ISIT), International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE Global Conference on Signal and Information Processing (GlobalSIP), DeepMath, MLSP Workshop

Journal Editor & Reviewer

Action Editor for Transaction on Machine Learning Research (TMLR)

Reviewer for Annals of Statistics, Information and Inference, SIAM Journal on Imaging Sciences (SIIMS), SIAM Journal on Optimization (SIOPT), Mathematical Programming (MP), Applied and Computational Harmonic Analysis (ACHA), Journal of Machine Learning Research (JMLR), Operations Research, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Signal Processing Magazine, IEEE Transactions on Pattern Recognition and Machine Intelligence (TPAMI), IEEE Transactions on Information Theory (TIT), IEEE Transactions on Image Processing (TIP), IEEE Transactions on Signal Processing (TSP), IEEE Transactions on Computational Imaging (TCI), IEEE Transactions on Geoscience and Remote Sensing (TGRS), IEEE Trans. on Circuit and System for Video Technology (TCSVT), IEEE Journal of Selected Topics on Signal Processing (JSTSP), IEEE Signal Processing Letters (SPL), Signal Processing (SP), Neurocomputing, Pattern Recognition (PR), PLOS One

Government Grant Panelist

National Science Foundation (NSF) CISE panelist

MENTORSHIP

Ph.D. Students

Lianghe Shi (Ph.D candidate at University of Michigan) Fall 2024 - present

Yixuan Jia (Ph.D candidate at University of Michigan) Summer 2024 - present

Zekai Zhang (Ph.D candidate at University of Michigan) Summer 2023 - present

Alec Xu (Ph.D candidate at University of Michigan) Fall 2023 - present

Soo-Min Kwon (Ph.D candidate at University of Michigan) Fall 2022 - present

Huijie Zhang (Ph.D candidate at University of Michigan) Fall 2022 - present

Siyi Chen (Ph.D candidate at University of Michigan) Fall 2022 - present

Xiang Li (Ph.D candidate at University of Michigan) Winter 2022 - present

Can Yaras (Ph.D candidate at University of Michigan) Fall 2021 - present

Xiao Li (Ph.D candidate at University of Michigan) Fall 2019 - present

Sheng Liu (Ph.D. New York University, now postdoc at Stanford) Fall 2019 - Summer 2023

Yuexiang Zhai (M.S. at Columbia University, Ph.D. at UC Berkeley) Spring 2018 - Winter 2021

Xiao Li (Ph.D. CUHK EE, now faculty at CUHK SZ) Spring 2019 - Summer 2020

Postdocs

Ismail Alkhouri (Ph.D. University of Central Florida) Summer 2023 - present

Peng Wang (Ph.D. CUHK) Winter 2022 - present

Yutong Wang (Ph.D. UMich, now Assistant Professor at IIT) Summer 2022 - Summer 2024

Dogyoon Song (Ph.D. MIT, now Assistant Professor at UC Davis) Summer 2022 - Summer 2024

Undergraduate/Master Students

Meng Wu (M.S. at University of Michigan) Fall 2024 - present

Wenda Li (B.S. at University of Michigan) Fall 2023 - present

Minzhe Guo (B.S. at University of Michigan, now PhD at UC Davis) Summer 2023 - Summer 2024

Yifu Lu (B.S. at University of Michigan) Winter 2023 - present

Jiyi Chen (M.S. at University of Michigan, noew PhD at IIT) Winter 2023 - Summer 2024

Saaketh Medepalli (B.S. at University of Michigan) Fall 2021 - Fall 2022

Jiahao Qiu (B.S. at University of Michigan, now Ph.D. at Princeton) Fall 2021 - Summer 2022

Xinyu Lu (B.S. University of Michigan, now M.S. at CMU) Fall 2021 - Summer 2022

Zhexin Wu (M.S. at University of Michigan, Now at ETH) Winter 2021-Summer 2021

PhD Dissertation Committees

Mingchen Li (adviser: Prof. Samet Oymak)	2024
Zeyun Sun (adviser: Prof. Alfred Hero)	2024
Bowen Liu (adviser: Prof. Hun-Seok Kim)	2024
Zongyu Li (adviser: Prof. Jeff Fessler)	Dec. 2023
Jiachen Sun (adviser: Prof. Morley Mao)	Oct. 2023
Haonan Zhu (adviser: Prof. Alfred Hero)	Aug. 2023
Guanhua Wang (adviser: Prof. Jeff Fessler & Douglas Noll)	May 2023
Sheng Liu (adviser: Prof. Carlos Fernandez-Granda)	Apr. 2023
Tuba Sarwar (adviser: Prof. Pei-Cheng Ku)	Mar. 2023
Kuan Han (adviser: Prof. Zhongming Liu)	Feb. 2023
Caroline Crockett (adviser: Prof. Jeff Fessler & Cindy Finelli)	Jul. 2022
Yutong Wang (adviser: Prof. Clayton Scott)	Jul. 2022